IH Used Paper BTECH-13-10-20

R13

Code No: 115AP

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, October - 2020
COMPILER DESIGN
(Computer Science and Engineering)

Time: 2 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Write the need of dividing compilation process into various phases.
b) What is top down parsing? Explain difficulties arise in top down parsing. [7+8]

2.a) Discuss about error handling mechanism in lexical analyzer.
b) Write predicative parsing algorithm and construct parsing table the input **id+id*id** for the following grammar. [7+8]
E→E+T/T, T→T*F/F, F→(E)/id.

3.a) Write LR parsing algorithm.
b) Check the following grammar is SLR or not.
S→L=R S→R L→*R/id R→L [7+8]

4.a) Explain the steps in SLR parser construction.
b) What is ambiguous grammar? How do you eliminate ambiguity in grammar? [7+8]

5.a) Explain heap storage allocation strategy.
b) Explain three address codes and illustrate with examples. [7+8]

6. Explain different data structures used for constructing symbol table. [15]

7.a) Explain loop optimization techniques and illustrate with examples.
b) Compare reducible Vs non reducible flow graphs. [7+8]

8.a) Construct DAG for a sub expression.
b) Write the procedure for generating code from DAG and explain with example. [7+8]

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